USN [ ] [23AS32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

FLUID MECHANICS

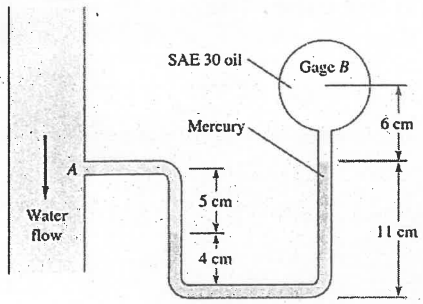
Time: 3 Hours **Maximum Marks: 100**

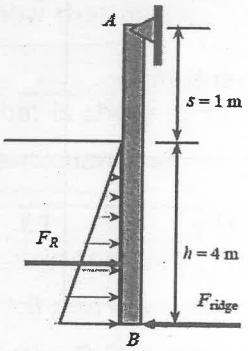
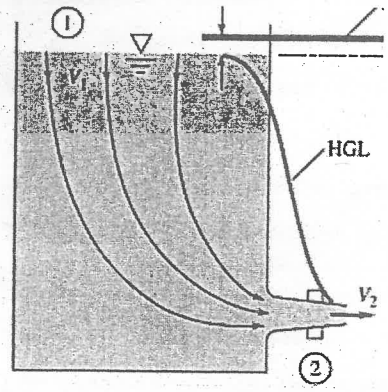
Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART-A			**5 x2=10**
1.	**a.**	Explain the term Compressibility.	L2	CO1	**02**
	b.	Explain the term Dynamic viscosity.	L2	CO1	**02**
	c.	Explain the term Uniform flow.	L2	CO2	**02**
	d.	Explain the term Fluid translation.	L2	CO2	**02**
	e.	Explain the term Fully developed flow.	L2	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Define pressure. Show that pressure at a point is same in all directions.	L2	CO1	**02** **03**
		OR			
	b.	What is shear stress. Explain Newtonian viscosity with examples for Newtonian and non-Newtonian fluids.	L2	CO1	**05**
3.	**a.**	Explain the reasons for surface tension in liquid. A 0.6 mm diameter glass tube inserted into water in a cup which is at 20 deg C. Determine the capillary rise of water in the tube. The surface tension of water is 0.073 N/m.	L2 L3	CO1 CO1	**2.5** **2.5**
		OR			
	b.	Distinguish between: **i.** Absolute pressure **ii.** Gauge pressure **iii.** Gauge vacuum and **iv.** Atmospheric Pressure with a neat sketch Explain with diagram that pressure at a point is same in all direction.	L2	CO1	**04** **01**
4.	**a.**	Explain hydrostatic law. Determine the gage pressure and absolute pressure at the bottom of tank containing water. The depth of water is 0.9 m.	L2 L3	CO2 CO2	**02** **03**
		OR			

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagara-112

	b.	Explain stability of a floating body. What is Meta centric height and under what condition a floating body is said to be stable, unstable and neutrally stable?	L2	CO2	02 03
5.	a.	Write the Bernoulli's equation and explain the different terms present in it. A Pitot-static probe is used to measure the speed of an aircraft flying at 1000 m. If the differential pressure reading is 3 kPa, determine the speed of the aircraft. Take air density as 0.91 kg/m^3.	L2 L3	CO3 CO3	02 03
		OR			
	b.	Explain the different types of forces acting on a fluid element. Write the Navier Stokes equation; explain the various terms.	L2	CO3	05
6.	a.	Explain Buckingham pi theorem and its benefits	L2	CO4	05
		OR			
	b.	What are geometric similarity and kinematic similarity?	L2	CO4	05
7.	a.	Define boundary layer thickness and velocity profile.	L2	CO5	05
		OR			
	b.	Explain Prandtls boundary layer concept and its benefits.	L2	CO5	05
		PART-C		**6 X 10=60**	
8.	a.	Explain how manometers are used to measure pressure at a point in fluid. Pressure gage *B* is to measure the pressure at point *A* in a water flow. If the pressure at *B* is 87 kPa, estimate the pressure at *A* in kPa. Assume all fluids are at 20C and use water density as 1000 kg/m^3. See Fig. given below-  	L2 L3	CO1	04 06
		OR			
	b.	Prove that the force due to buoyancy is equal to the weight of the fluid displaced. A standard basket ball (mass 0.62 kg and diameter 24.3 cm.) is held fully under water. Calculate the buoyant force acting on the ball. If the ball is released, determine what portion of it will be under water?	L2 L3	CO2	05 05

9.	**a.**	Define or explain **i.** Control volume **ii.** Capillarity **iii.** Local acceleration **iv.** Dimensional variables **v.** Major losses	L2	CO1 CO2 CO3 CO4	10
		OR			
	b.	Define cohesive force and adhesive force. Explain the reason for viscosity of a fluid. What is the effect of temperature on viscosity of air and water?	L2	CO1	**04** **06**
10.	**a.**	Explain the reason for surface tension in a fluid. Obtain the expression for pressure difference for a spherical droplet. Find the pressure difference for a water bubble of radius 0.5 mm. The surface tension for water is 0.073 N/m.	L2 L3	CO1	**06** **04**
		OR			
	b.	Derive an expression for centre of pressure on a submerged plate inclined to the surface of the fluid. With reference to Fig. shown below, determine the horizontal force acting on the wall. The fluid is assumed to be water.  	L2 L3	CO1	**05** **05**
11.	**a.**	Refer Fig 1 given below. Obtain the expression for velocity at the station 2.  A closed tank has an orifice of diameter 0.025 m in one of its vertical sides. The height of water in the tank is 0.61 m above the centerline	L2 L3	CO3	**05** **05**

		of the orifice and the air pressure inside the tank is 13780 pa above atmosphere. Calculate the water discharge rate from the orifice, assuming that the discharge co-efficient is 0.8.			
		OR			
	b.	The Euler equation along a streamline is given by $VdV + dp/\rho + gdz = 0$. Obtain Bernoullies equation using above equation. Explain in detail, the significance of the different terms appearing in the Bernoullis equation.	L2	CO3	**05** **05**
12.	**a.**	Explain major and minor losses for fluid flow through pipes with examples. Describe the steps involved in dimensional analysis based on Buckingham pi theorem.	L2	CO4	**05** **05**
		OR			
	b.	An oil of viscosity 10 poise, flows between two parallel fixed plates, which are kept at a distance of 50 mm. Find the rate of oil flow between the plates if the pressure drop 0.3 N/cm^2 over a length of 1.2 m. The width of plate is 200 mm.	L3	CO4	**10**
13.	**a.**	Define boundary layer and, boundary layer thicknes and boundary layer energy thickness. Obtain the expression for displacement thickness and momentum thickness.	L2	CO5	**04** **06**
		OR			
	b.	If the velocity profile is given by $u/U_\infty = 1-(y/\delta)^{1/5}$, find out the displacement thickness and momentum thickness.	L3	CO5	**10**